Azure Virtual Machine with a GPU

Tutorial: how to login & use it

1. Email

You should receive an email with the invite to join the lab. Click the blue button.

Microsoft Azure

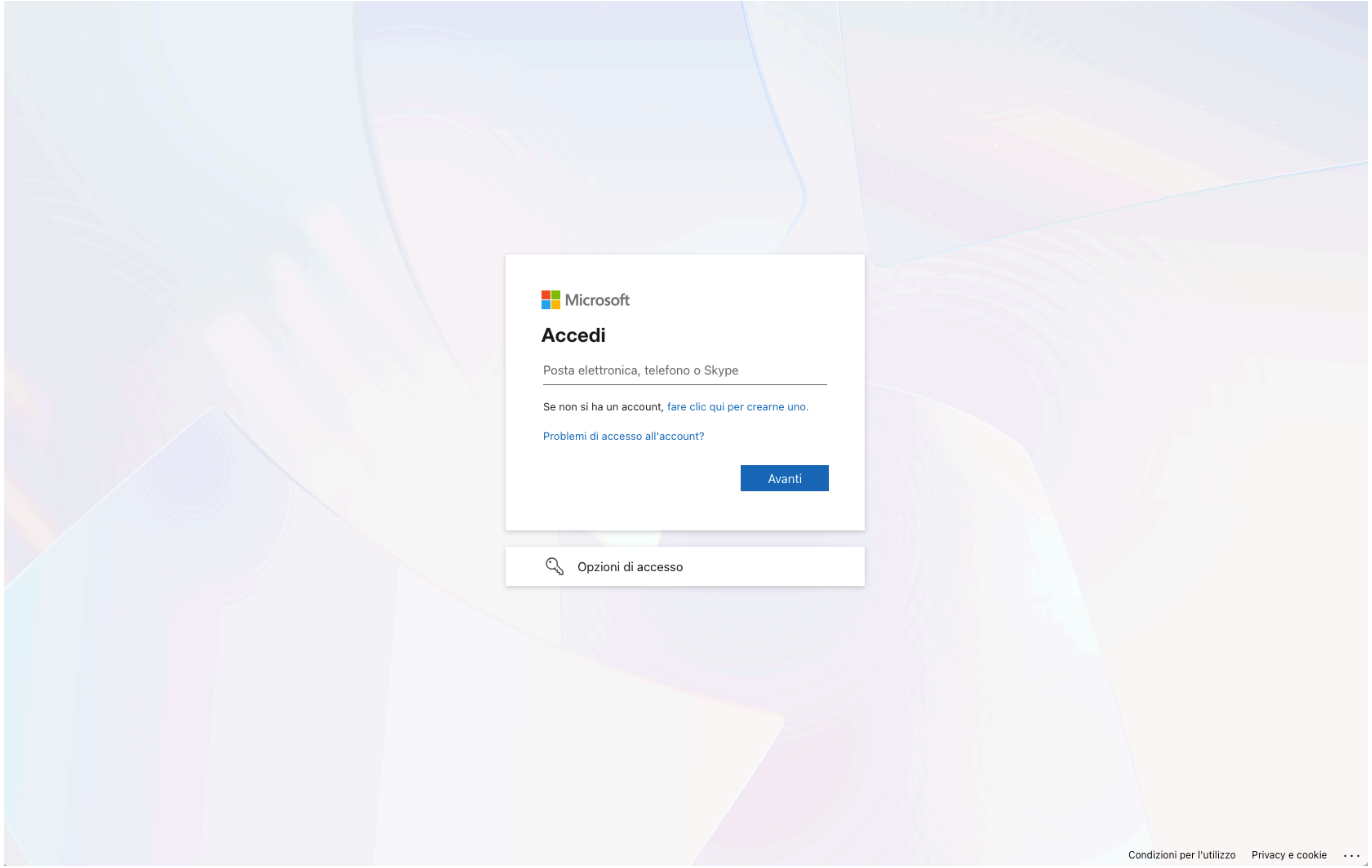
mad-supertomasi@unitrento365.onmicrosoft.com ti ha invitato al lab: DeepLearning-Ricci

Iscriviti subito per accedere alle macchine virtuali nel lab.

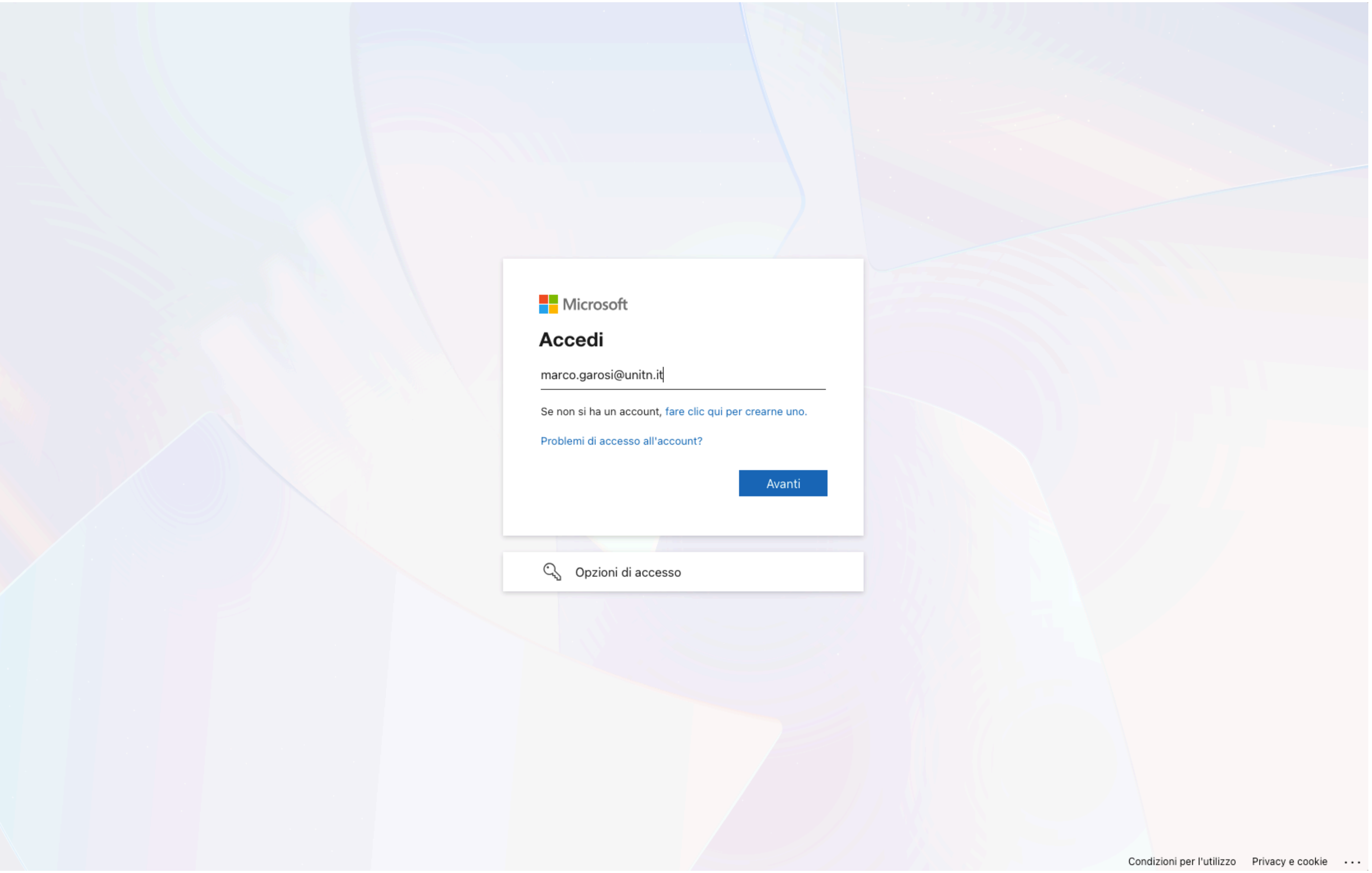
Registrati al lab >

2. Login

You will be redirected to this login page.
Use your @studenti.unitn.it account! (See next slide)



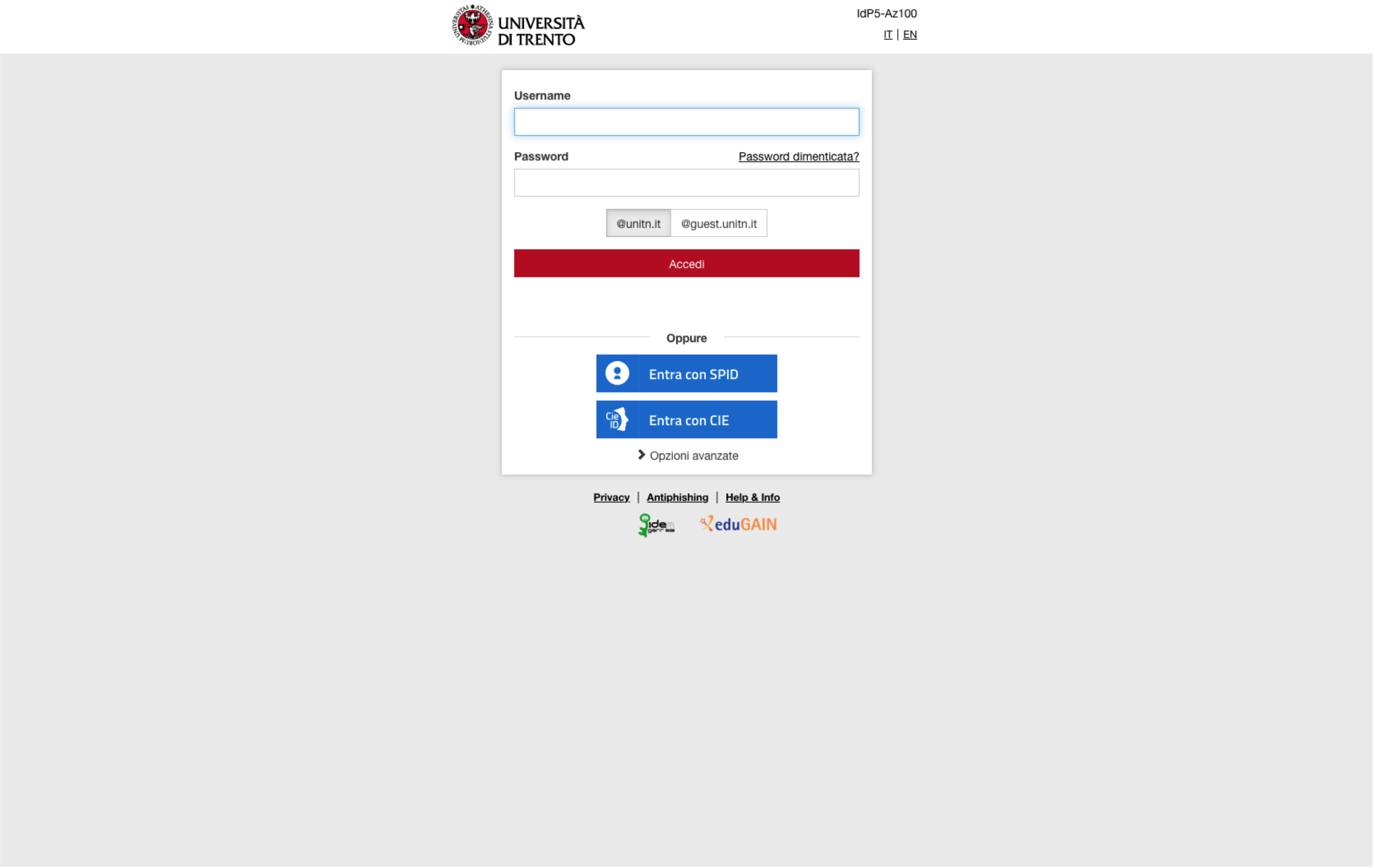
2. Login



Use your @studenti.unitn.it account!
Simply enter your email and hit the blue button.

Note: only the student who received the email can do this!

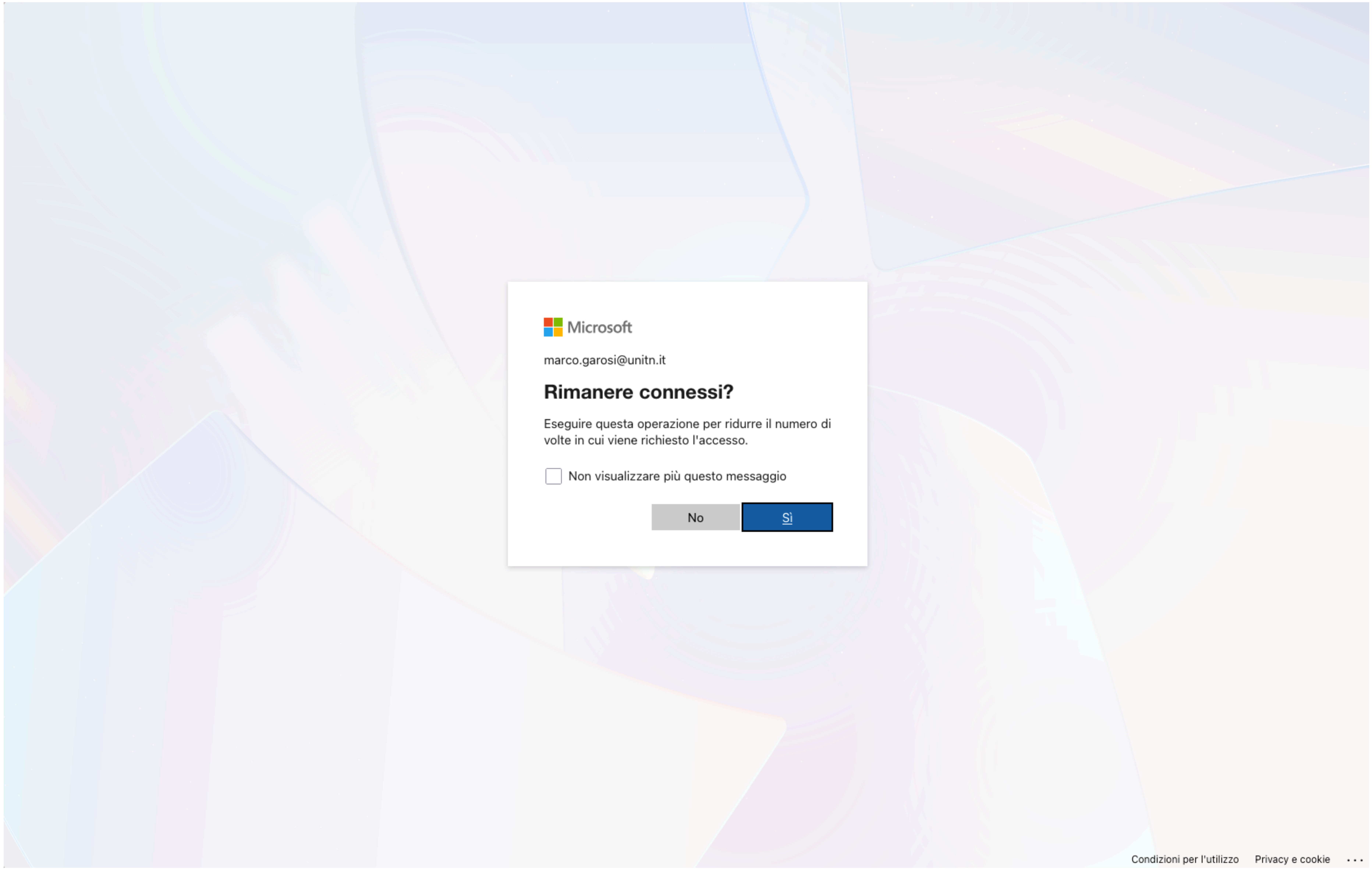
2. Login



After hitting the blue button in the previous page, you will be redirected to the usual UniTN login page.
Log in using your data, and hit the red button.

2. Login

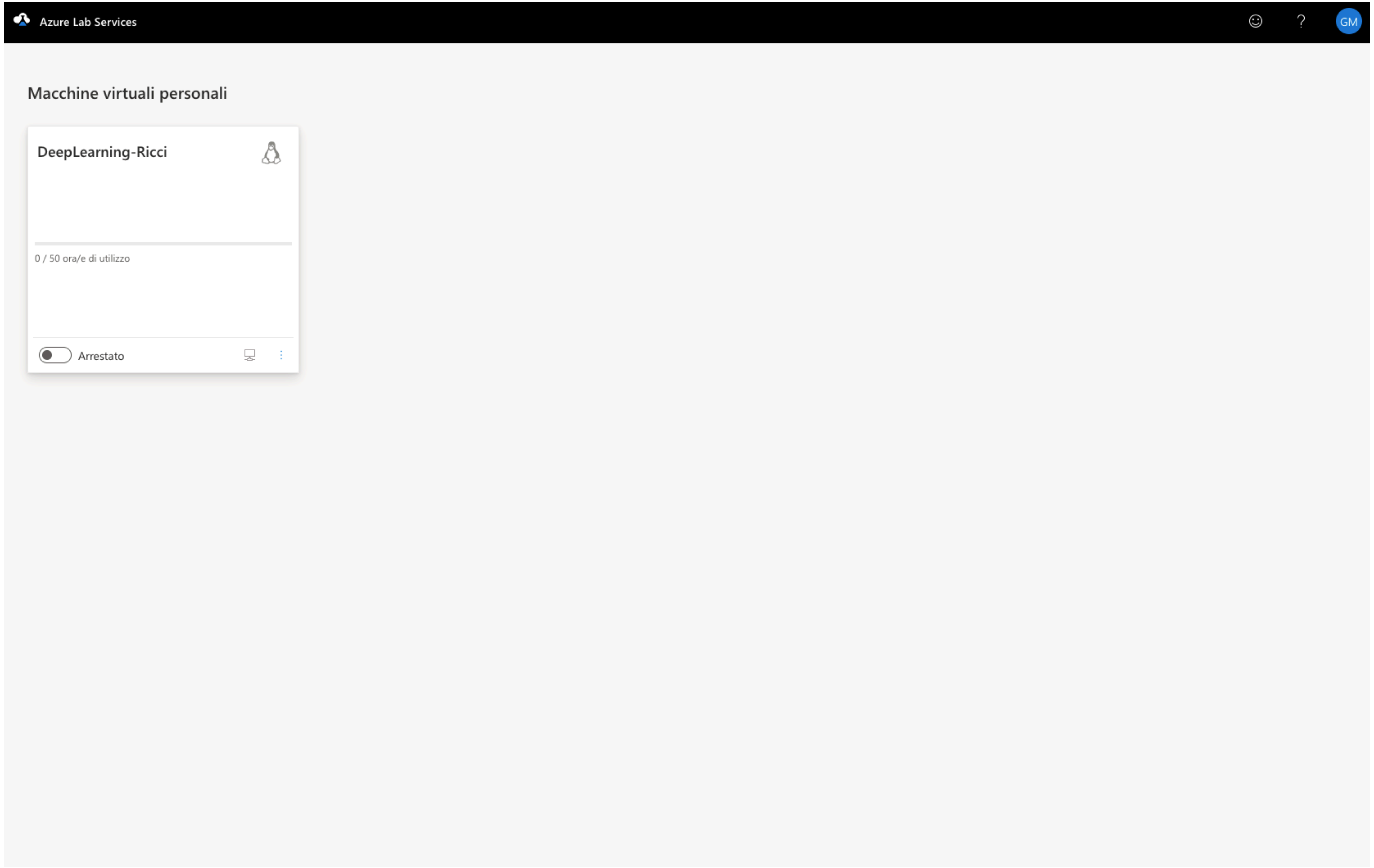
You will be redirected to Azure's login page. You can choose to remain connected to the account by hitting "Yes" (blue button).



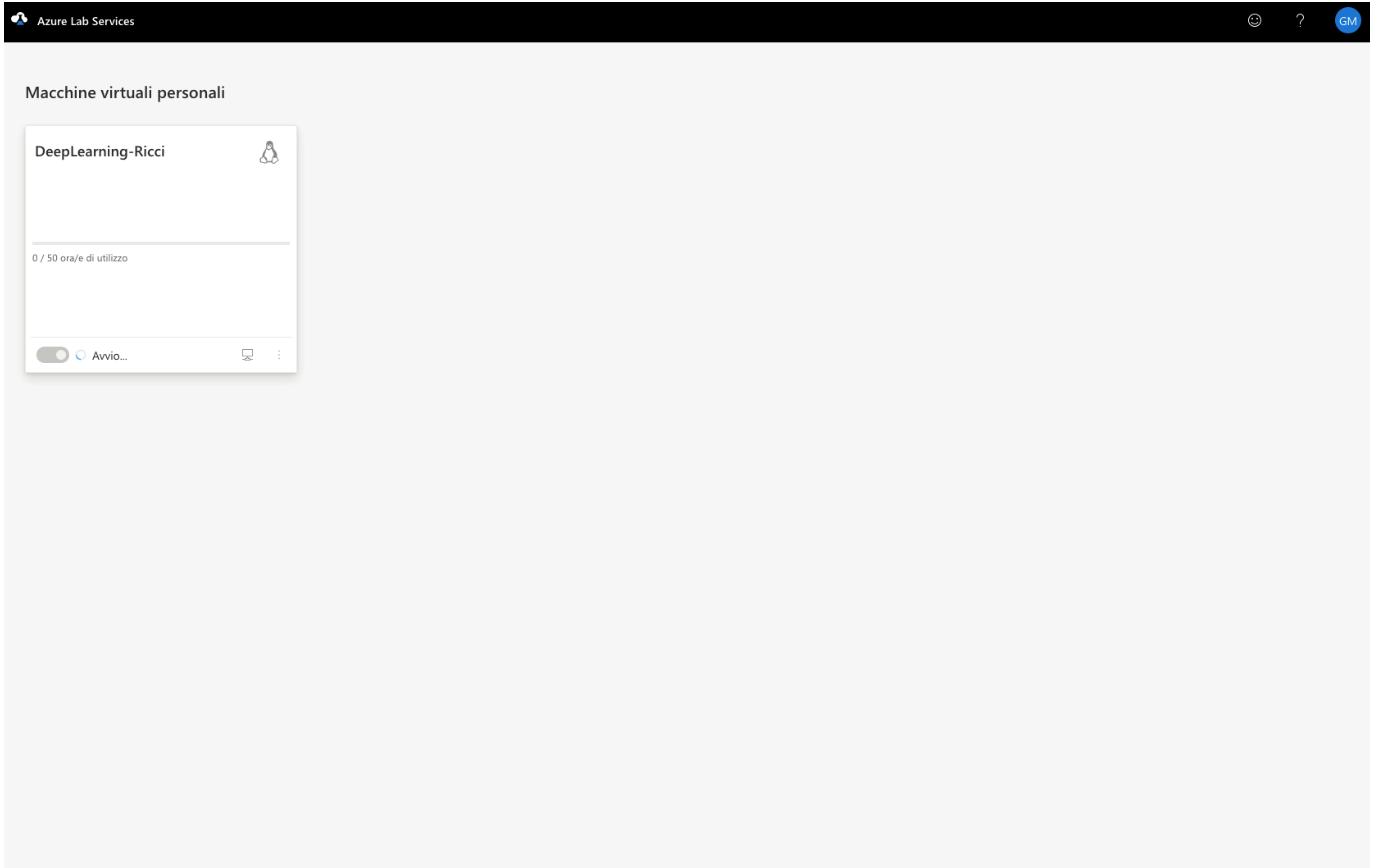
3. Dashboard

You are finally in the dashboard. You should see something like this. This is your virtual machine, and you can see how many hours you've used from your budget.

Note: remember to always turn off the VM when you are done using it, otherwise you will waste hours!



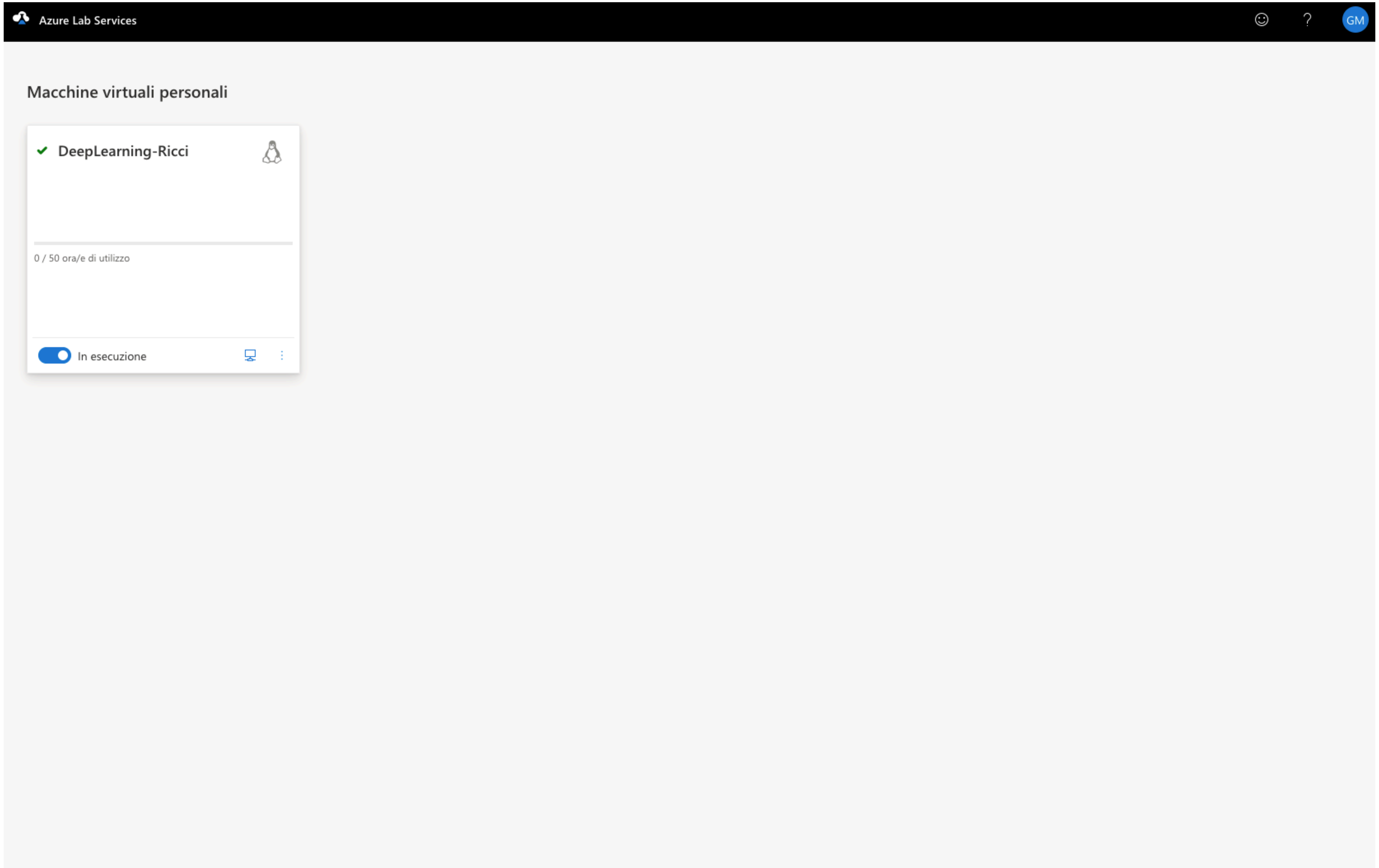
4. Start the VM



Click the button on the bottom-left side of the virtual machine's card. That will boot the VM.

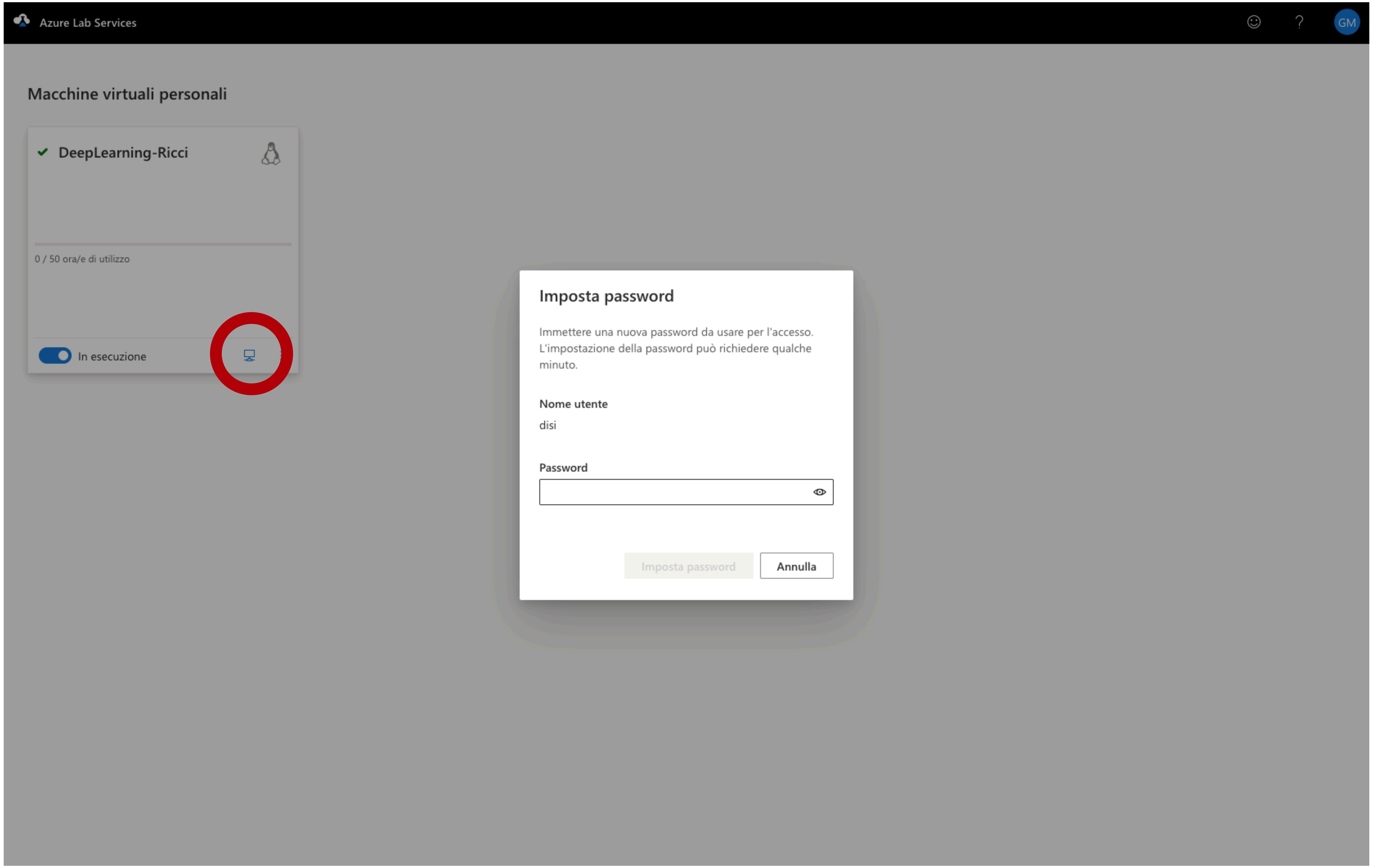
Note: it might takes a while to do that! Be patient and don't refresh the page.

4. Start the VM



Once the machine is booted, the slider will say “In esecuzione” or “Running”. The VM is ready.

5. Set a password

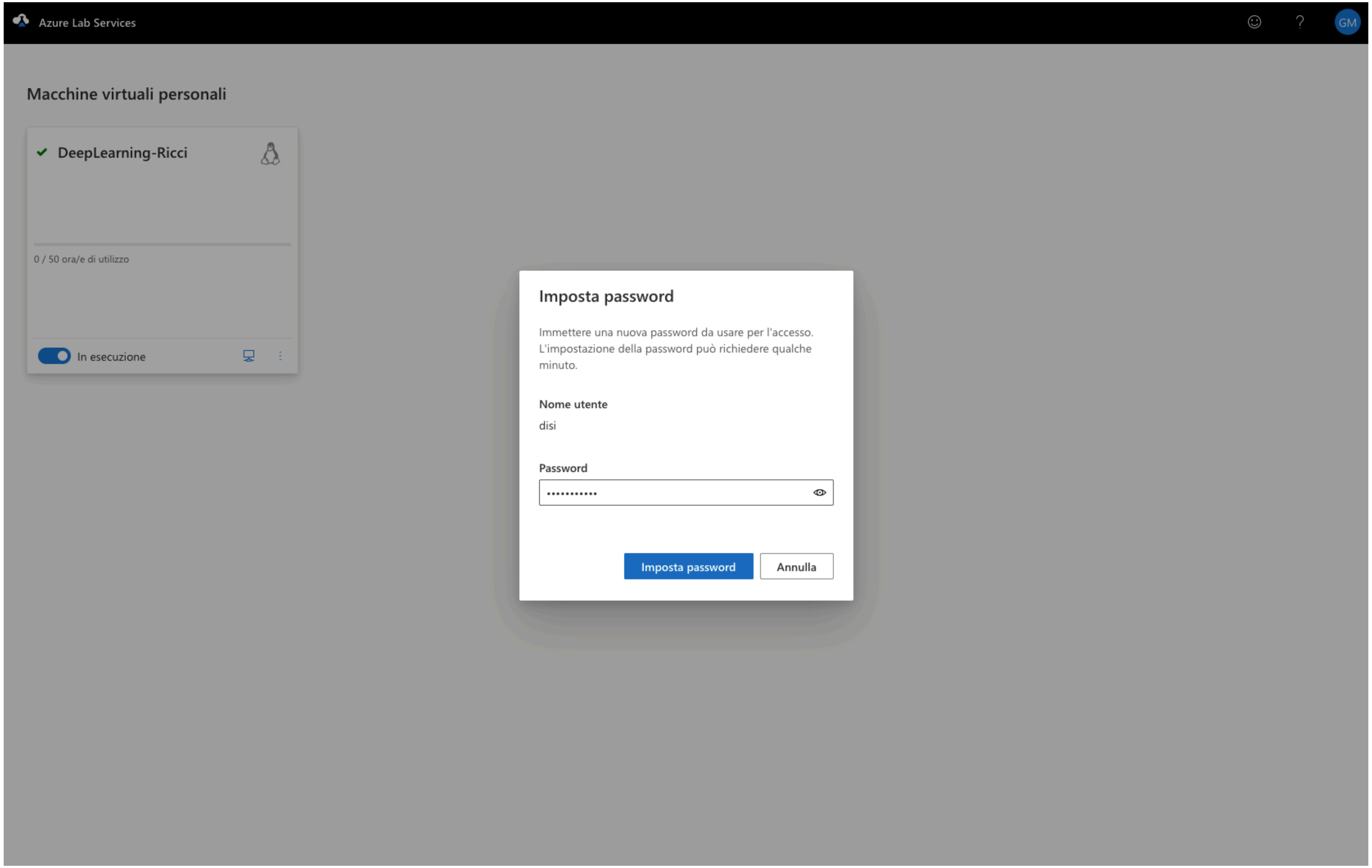


You can log into the VM via SSH. However, you need to set a password. You can do so by clicking the circled icon. It will open the popup shown here.

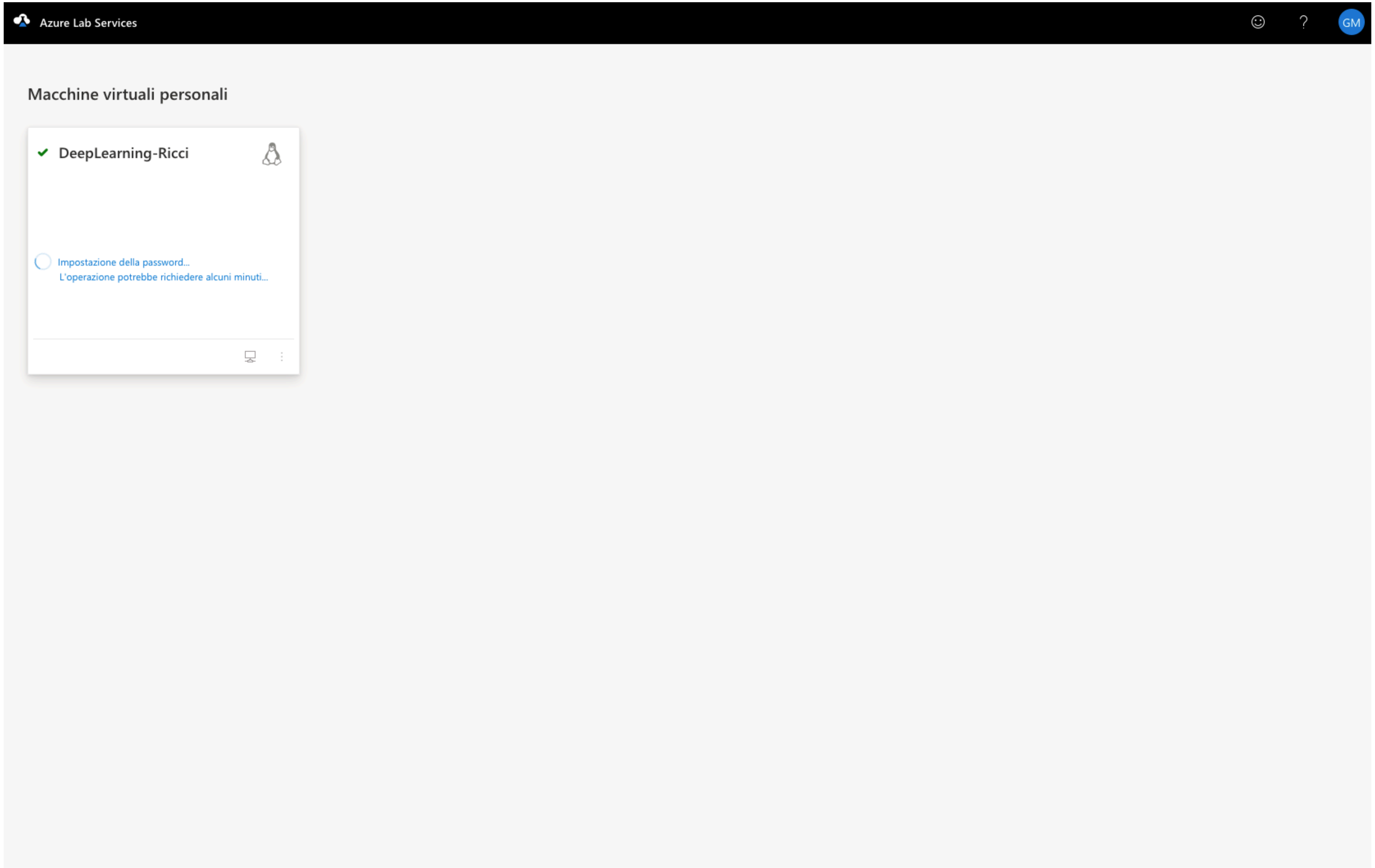
5. Set a password

Just choose a password of your choice. You can always reset it. You can share it with your teammates, so that they can also log into the VM.
Hit the blue button when you've set the password.

Note: if the VM is off, you can't log into it. So the "admin" will have to turn on the VM for the teammates, too.

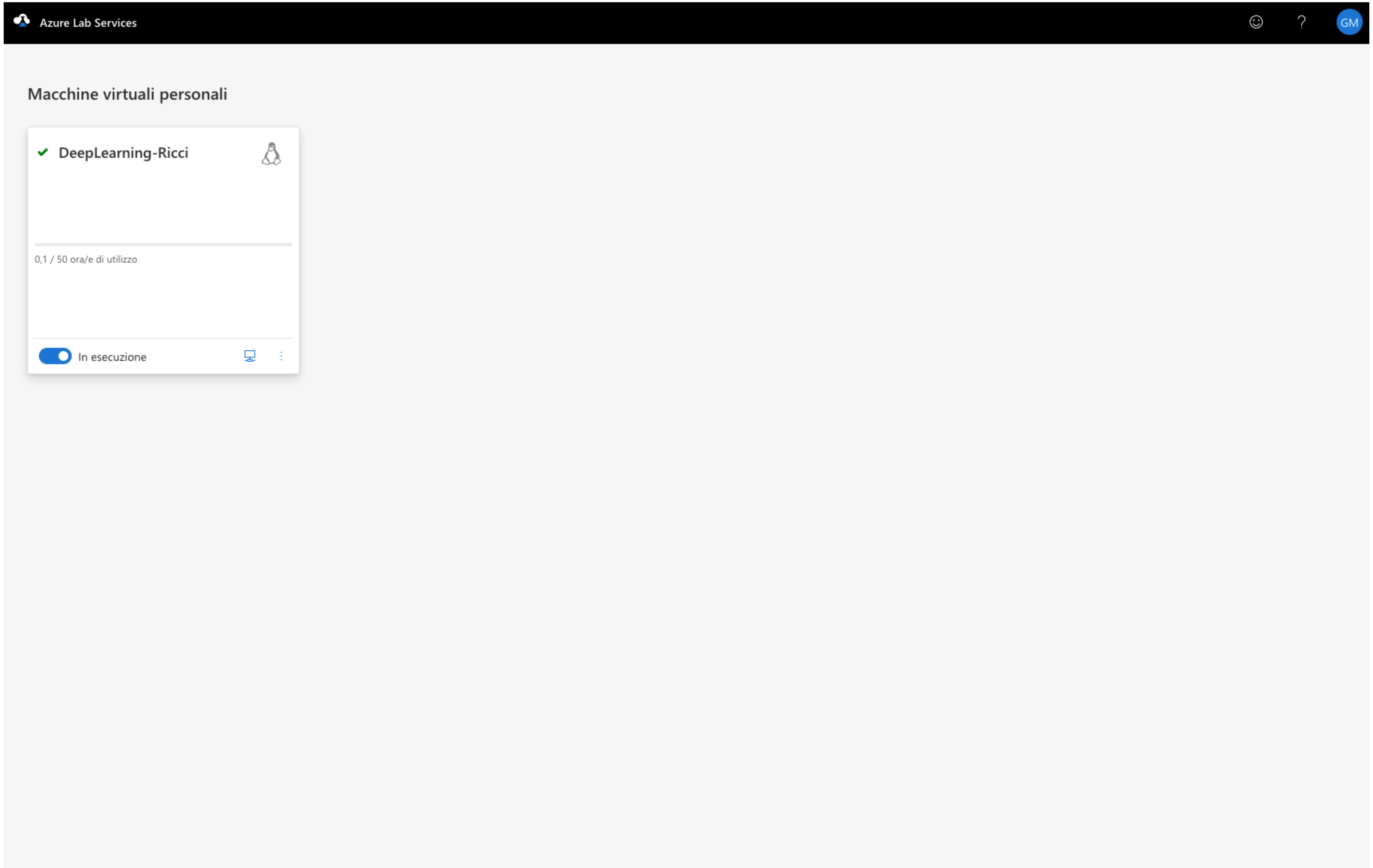


5. Set a password



Wait a couple of minutes so that the password is correctly set. Don't refresh the page.

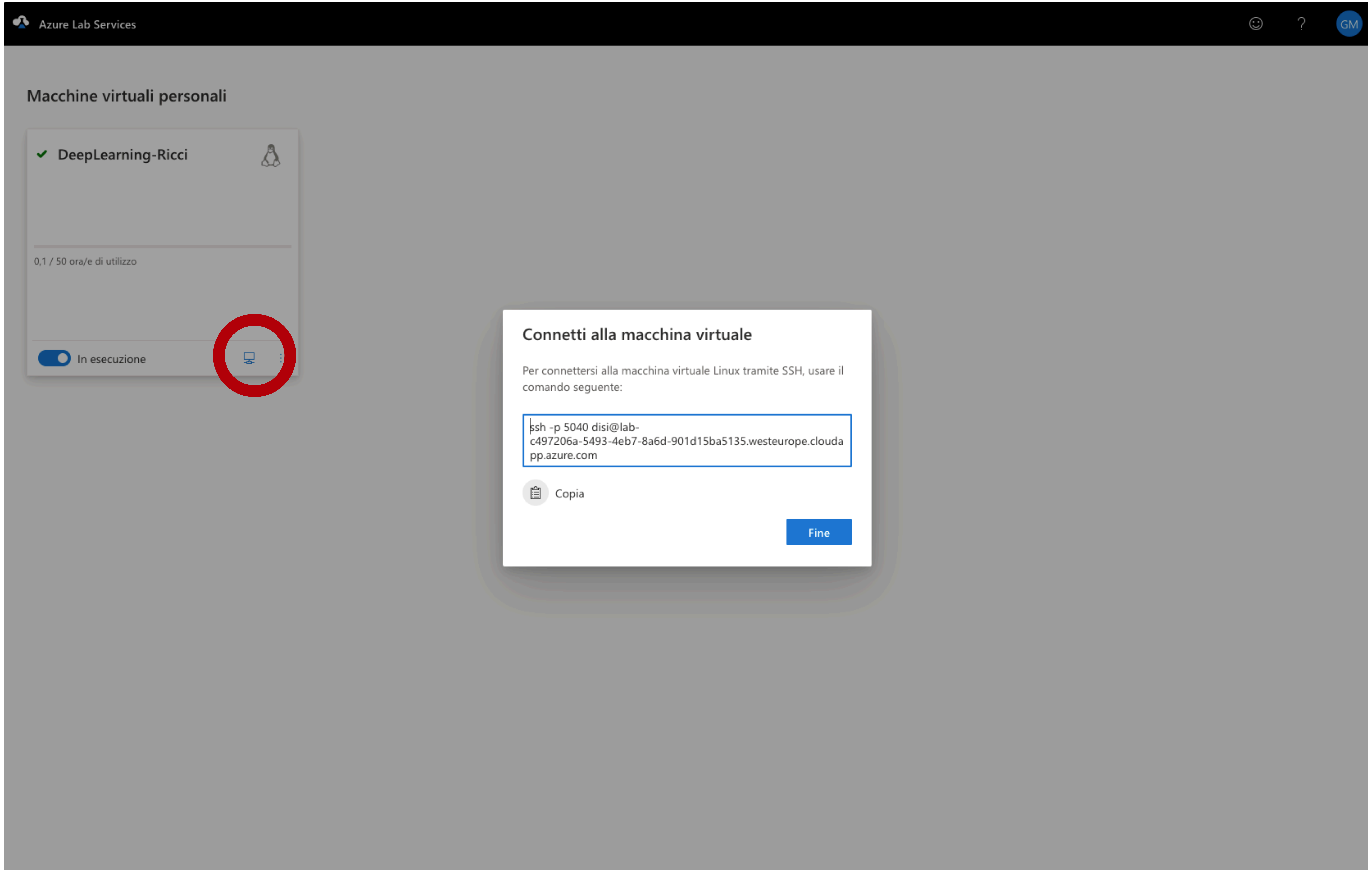
5. Set a password



Once it’s done, you should see something like this. The initial setup is completed, and you can now log in into the VM.

6. Login

If you click again the icon in the red circle, it should now show you the address of your machine. It provides you the command to connect via SSH. **Copy-paste it.**



6. Login

Move to your terminal and paste the command. Then, hit Enter.

6. Login

If something like this happens, just type in “yes” and hit Enter.

```
➜  ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
The authenticity of host '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040 ([52.157.80.239]:5040)' can't be established.
ED25519 key fingerprint is: SHA256:lLn8huPhj/5EVXc094Qlg1hmkvbBfiQkvv98ioTt028
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

6. Login

Type in your password (the one you set earlier in Azure's web console) and hit Enter.

Note: the password will not appear, and you will not see any blinking indicator (as it usually happens in the terminal). Just type the password and hit Enter, your keyboard is working!

```
➜  ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
The authenticity of host '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040 ([52.157.80.239]:5040)' can't be established.
ED25519 key fingerprint is: SHA256:lLn8huPhj/5EVXc094Qlg1hmkvbBfiQkvv98ioTt028
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040' (ED25519) to the list of known hosts.
disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com's password:
```

6. Login

If you did everything correctly, you should see something like this. You have eventually made it to the VM!
You are an admin to this VM, so you can do (almost) whatever you want with it.

You can test the GPU with `nvidia-smi`, which will show you some info about the GPU.

```
➜  ~ ssh -p 5040 disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com
The authenticity of host '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040 ([52.157.80.239]:5040)' can't be established.
ED25519 key fingerprint is: SHA256:lLn8huPhj/5EVXc094Qlg1hmkvbBfiQkvv98ioTt028
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com]:5040' (ED25519) to the list of known hosts.
disi@lab-c497206a-5493-4eb7-8a6d-901d15ba5135.westeurope.cloudapp.azure.com's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Apr 23 10:56:10 UTC 2026

  System load:  0.08              Processes:             166
  Usage of /:   14.3% of 122.94GB Users logged in:       0
  Memory usage: 0%                IPv4 address for eth0: 10.0.0.44
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

18 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

disi@labVDMJXW:~$
```

6. Login

`nvidia-smi` shows something like this:

```
disi@labT31YWB:~$ nvidia-smi
Thu Apr 23 12:19:13 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 555.42.06              Driver Version: 555.42.06      CUDA Version: 12.5     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla V100-PCIE-16GB           Off |   00000001:00:00.0 Off |                  Off |
| N/A   30C    P0             25W /  250W |       1MiB /  16384MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                              GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

7. Environment setup

Now you need to install Python and PyTorch.
Using uv will make this easy.

Just type in the following commands:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
source $HOME/.local/bin/env
mkdir <your_project_folder>
cd <your_project_folder>
uv init --python 3.12
uv venv
source .venv/bin/activate
uv pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126
```

7. Environment setup

This is the output that you should expect from the commands in the previous slide.

```
disi@labT31YWB:~$ curl -LsSf https://astral.sh/uv/install.sh | sh
downloading uv 0.11.7 x86_64-unknown-linux-gnu
installing to /home/disi/.local/bin
  uv
  uvx
everything's installed!

To add $HOME/.local/bin to your PATH, either restart your shell or run:

    source $HOME/.local/bin/env (sh, bash, zsh)
    source $HOME/.local/bin/env.fish (fish)
disi@labT31YWB:~$ source $HOME/.local/bin/env
disi@labT31YWB:~$ mkdir test
disi@labT31YWB:~$ cd test/
disi@labT31YWB:~/test$ uv init --python 3.12
Initialized project `test`
disi@labT31YWB:~/test$ uv venv
Using CPython 3.12.13
Creating virtual environment at: .venv
Activate with: source .venv/bin/activate
disi@labT31YWB:~/test$ source .venv/bin/activate
(test) disi@labT31YWB:~/test$ uv pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126
Resolved 32 packages in 1.64s
Prepared 32 packages in 27.91s
Installed 32 packages in 264ms
 + cuda-bindings==12.9.4
 + cuda-pathfinder==1.2.2
 + cuda-toolkit==12.6.3
 + filelock==3.25.2
 + fsspec==2026.2.0
 + jinja2==3.1.6
 + markupsafe==3.0.3
 + mpmath==1.3.0
 + networkx==3.6.1
 + numpy==2.4.3
 + nvidia-cublas-cu12==12.6.4.1
 + nvidia-cuda-cupti-cu12==12.6.80
 + nvidia-cuda-nvrtc-cu12==12.6.85
 + nvidia-cuda-runtime-cu12==12.6.77
 + nvidia-cudnn-cu12==9.10.2.21
 + nvidia-cufft-cu12==11.3.0.4
 + nvidia-cufile-cu12==1.11.1.6
 + nvidia-curand-cu12==10.3.7.77
 + nvidia-cusolver-cu12==11.7.1.2
 + nvidia-cusparse-cu12==12.5.4.2
 + nvidia-cusparselt-cu12==0.7.1
 + nvidia-nccl-cu12==2.28.9
 + nvidia-nvjitlink-cu12==12.6.85
 + nvidia-nvshmem-cu12==3.4.5
 + nvidia-nvtx-cu12==12.6.77
 + pillow==12.1.1
 + setuptools==70.2.0
 + sympy==1.14.0
 + torch==2.11.0+cu126
 + torchvision==0.26.0+cu126
 + triton==3.6.0
 + typing-extensions==4.15.0
```

8. Test the installation

Now you can test the GPU (and if PyTorch recognizes it) by running Python in interactive mode, as shown here.
No errors: everything is working fine.

```
(test) disi@labT31YWB:~/test$ python
Python 3.12.13 (main, Apr 14 2026, 14:29:00) [Clang 22.1.3 ] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> torch.cuda.is_available()
True
>>> x = torch.zeros((100, 100), device="cuda")
>>> x.device
device(type='cuda', index=0)
>>> x ** 2
tensor([[0., 0., 0.,  ..., 0., 0., 0.],
        [0., 0., 0.,  ..., 0., 0., 0.],
        [0., 0., 0.,  ..., 0., 0., 0.],
        ...,
        [0., 0., 0.,  ..., 0., 0., 0.],
        [0., 0., 0.,  ..., 0., 0., 0.],
        [0., 0., 0.,  ..., 0., 0., 0.]], device='cuda:0')
>>> exit()
```

Turn off the VM!

Just a final reminder to turn off the VM when you're not using it, otherwise you will run out of budget pretty quickly.